

Information-Flow Investigation

Why the trained decoder-only Transformer has data but cannot use it reliably during generation

Scope: tokenizer, data construction, model, loss, gradients, checkpoints, SFT, and inference.

Constraints: non-destructive investigation; no retraining, new data, code changes, or Exp015.

Project: D:\project\LLM\LLM

Executive conclusion

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The primary failure is not a broken tensor pipeline or decoding algorithm. The checkpoints learn local token statistics, but do not assign high probability to semantically correct continuations. SFT then amplifies generic conversational templates because the Exp011 responses are highly duplicated. Exp014 also has a concrete duplicated dialogue-prefix defect.

For the controlled prompt `The capital of France is`, the correct `Paris` token ranked 1,824 in Exp012 base, 1,902 in Exp012 SFT, and 1,449 in Exp014 SFT. Generic tokens such as `a`, `the`, and `located` ranked much higher. This means the correct continuation is already low probability before decoding.

1. Information-flow map

Corpus

- > tokenizer_exp003
- > binary token stream / JSONL instruction examples
- > shifted training windows
- > Transformer forward pass
- > next-token cross-entropy
- > gradients and AdamW updates
- > checkpoint
- > prompt tokenization
- > forward pass and logits
- > greedy / top-k / top-p decoding
- > final output

The investigation found no primary failure in causal masking, optimizer stepping, checkpoint architecture loading, or pretraining target shifting.

2. Tokenizer findings

Implementation: `tokenizer/tokenizer.py`. Vocabulary size is 10,000 with valid `<PAD>`, `<UNK>`, `<BOS>`, and `<EOS>` tokens.

- Representative book, Wikipedia, factual, and conversational strings round-trip exactly through encode/decode.
- `€` is a real vocabulary token, not an unknown token.
- Actual Exp012 train split: 31,639,844 tokens, 97.62% vocabulary utilization, 0.426% `<UNK>`, 0.073% `€`.
- Actual validation split: 4,772,344 tokens, 94.92% utilization, 0.379% `<UNK>`, 0.078% `€`.

Assessment: tokenizer information loss exists but is too small to explain generic generation. The unusual `€` output is not proof of tokenizer corruption.

3. Data and target flow

Implementation: `training/dataset.py`. Pretraining loads `context_length + 1` tokens and constructs:

```
input  = tokens[start : start + context_length]
target = tokens[start + 1 : start + context_length + 1]
```

A real binary-window diagnostic confirmed the one-token shift exactly. BOS/EOS markers are added per source line and there is no pretraining padding.

SFT implementation: `training/instruct_train.py`. Prompt labels are `-100`; assistant response tokens are real targets. PyTorch cross-entropy ignores `-100` by default. The response-only loss is therefore correctly configured.

4. Model and attention findings

Implementations: `model/model.py`, `model/transformer.py`, and `model/attention.py`.

- Embedding, positional embedding, Q/K/V projections, head splitting, scaling, residuals, LayerNorm, FFN, final normalization, and LM head are structurally correct.
- Logits have shape `[B, T, V]`.
- The causal mask has shape `[1, 1, 256, 256]`, equals a lower-triangular matrix, and enables zero future-position connections.
- Checkpoint configuration is loaded from saved metadata, so inference architecture dimensions match training.

Assessment: no architecture or future-token leakage bug was found.

5. Loss and gradient findings

Loss is genuine next-token cross-entropy. A non-destructive backward pass on Exp012 produced finite, nonzero gradients:

Component	Gradient norm
Token embedding	0.0280
First Transformer block	0.0143
Middle block	0.0241
Last block	0.0157
LM head	0.4564

The training loop performs backward propagation, clipping, optimizer updates, scheduler stepping, AMP handling, and finite-loss checks. Gradient starvation is not supported by the evidence.

6. Pretraining behavior

Exp012 reached validation loss 2.6152 and perplexity 13.67. This demonstrates token-level compression, not semantic understanding.

Observed continuations include:

The capital of France is -> the largest city of the city of the city ...
In the beginning -> of the morning of the morning ...

The base model learned whitespace, punctuation, common word transitions, and some syntax. It did not demonstrate reliable factual association, long-range discourse control, or semantic conditioning.

7. SFT behavior

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The Exp011 report contains 10,000 training examples but only 400 unique normalized responses. Exact duplicate responses total 11,600 across the recorded splits; common responses repeat approximately 50 times.

This makes low SFT loss compatible with generic answers. The model can minimize loss by learning common response templates rather than learning a robust mapping from arbitrary prompts to appropriate responses.

Exp014 contains an additional formatting problem. Raw examples already contain dialogue labels, but preprocessing adds another prefix, producing inputs such as:

```
<BOS>User: User: How can I help you?  
Assistant: ...  
Assistant: Ok. ...
```

This is a real prompt-conditioning defect for Exp014.

8. Inference audit

Implementation: `experiments/inference/generate.py`. Checkpoint loading, tokenizer selection, BOS handling, context truncation, last-position logits, temperature, top-k, top-p, greedy selection, and EOS stopping are correct.

The API uses the intended `User: {message}\nAssistant:` format and loads the Exp012 SFT checkpoint by default. No silent model-architecture mismatch was found.

9. Root-cause candidates

Candidate	Evidence	Confidence
Weak semantic representation	Correct factual token ranks very low; context changes have limited semantic effect.	High
SFT template collapse	400 unique responses repeated heavily; generic phrases dominate outputs.	High
Exp014 formatting mismatch	Inputs contain duplicated <code>User: / Assistant:</code> prefixes.	High for Exp014
Tokenizer information loss	0.426% train <code><UNK></code> rate.	Low as main cause
Loss/target alignment failure	Direct shift and masking checks pass.	Very low
Causal attention or gradient failure	Mask is strictly causal; all tested gradients are finite and nonzero.	Very low
Decoding failure	Correct token is already low probability before decoding.	Very low

10. What should not change yet

- Do not change model size, tokenizer architecture, corpus scale, loss shifting, attention, checkpoint loading, or decoding parameters yet.
- Do not retrain until teacher-forced behavior is measured across a larger sample.

11. Next single diagnostic

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Run teacher-forced evaluation over hundreds of known continuations for Exp012 base, Exp012 SFT, and Exp014 SFT. For every example record:

```
prompt
expected next token
rank
probability
top-5 alternatives
```

This will quantify whether the remaining problem is factual knowledge absence, weak semantic conditioning, SFT template collapse, or generation-only degradation.

12. Validation notes

Focused dataset, attention, and API tests produced 7 passes and one unrelated compatibility failure: `tests/test_dataset.py` expects legacy attributes `bos_id / eos_id`, while the tokenizer exposes `bos_token_id / eos_token_id`. No project files were modified during this investigation.